7 **(a)** Tungsten, a transition metal, is commonly used as a target metal for the production of X-rays. The energy levels of the K- to M-shells for tungsten are shown in Fig. 7.1 below.

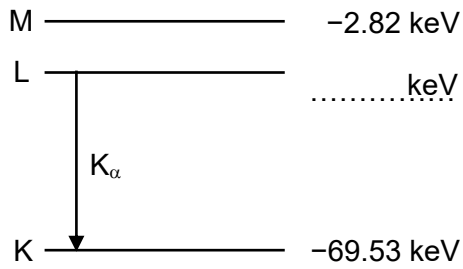

Fig. 7.1

The wavelength of the photon produced by the K_α transition is 21.65 pm.

(i) Complete Fig. 7.1 by filling in the energy level of the L-shell for tungsten.

[2]

(ii) Determine the wavelength of the photon produced by L_α transition.

wavelength = …… pm [1]

For Examiner's Use

The intensity of various photon wavelengths from electron bombardment of a tungsten target metal is shown in Fig. 7.2 below.

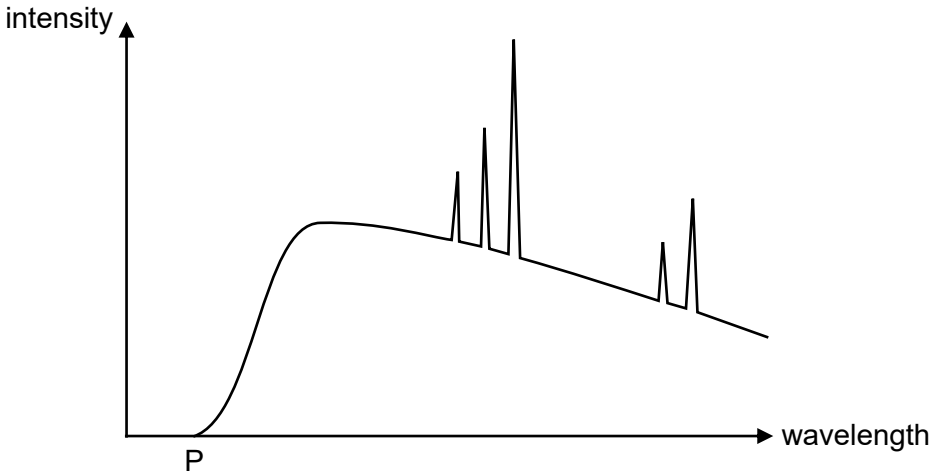

Fig. 7.2

(iii) On Fig. 7.2, label the peaks for K_α and L_α transitions. [2]

(iv) State the significance of the value of P.

.....

..... [1]

(v) With reference to Fig. 7.1, explain why the electrons bombarding the target metal must have a minimum energy of 69.53 keV to produce the spectrum shown in Fig. 7.2.

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..... [3]

For Examiner's Use

(b) Fig. 7.3 below shows a typical setup for producing such X-ray beams.

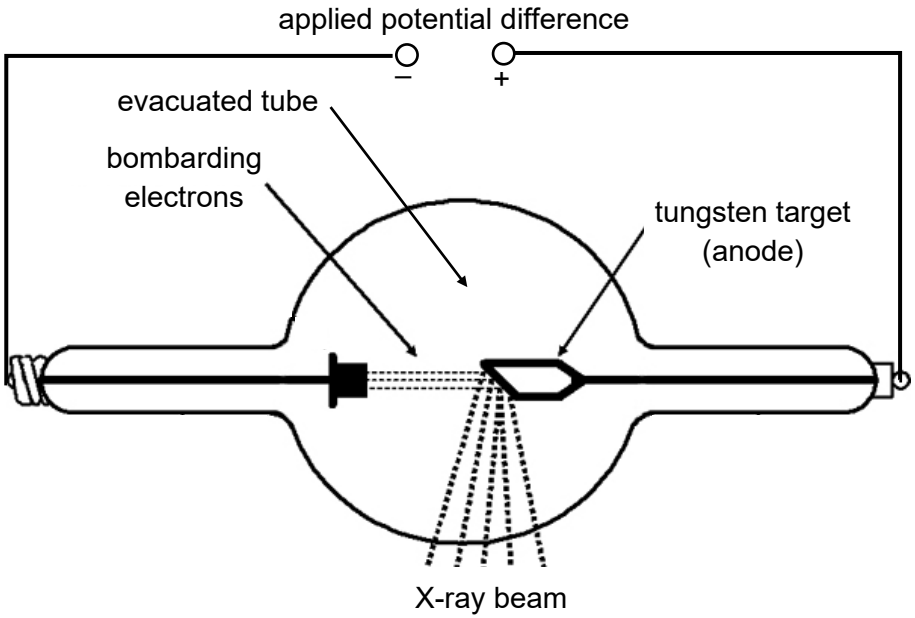

Fig. 7.3

For safety reasons, the wavelength of radiation used for medical X-rays is no shorter than 50 pm. Explain, with calculation, how the setup in Fig. 7.3 can be adjusted to ensure that the wavelength of the X-rays produced does not fall below 50 pm.

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..... [3]